

MedFace – Facial Recognition-Based Patient Record System

Product Requirements Document (PRD)

1. Overview

MedFace is a secure web-based platform designed for healthcare professionals to manage patient medical histories.

2. Goals and Objectives

- Enable doctors to retrieve a patient's medical history using face recognition.
- Maintain complete electronic medical records (EMRs).
- Securely store and manage patient data.
- Ensure role-based access for doctors, admins, and other staff.
- Reduce errors in patient identification.
- Create a scalable system for clinics or hospitals.

3. Target Users

- Doctors
- Nurses / Healthcare Workers
- Hospital Admins
- (Optional) Patients for self-service access

4. Key Features

- Patient Registration
- Facial Recognition
- Patient History Access
- Doctor Dashboard
- Admin Dashboard
- Authentication
- Security

5. Architecture

Frontend: React.js (Tailwind UI), Webcam integration

Backend: Node.js (Express) or Python (FastAPI)

Face Recognition: FaceNet or Dlib

Database: MongoDB or PostgreSQL

6. Functional Requirements

- FR-01: Secure login for doctors
- FR-02: Patient registration with face capture
- FR-03: Identify patients via webcam
- FR-04: Display patient medical history
- FR-05: Upload prescriptions and notes
- FR-06: Role-based access control
- FR-07: Record access logging

7. Non-Functional Requirements

- NFR-01: Face recognition < 3 seconds
- NFR-02: Data encryption at rest and in transit
- NFR-03: Uptime > 99%
- NFR-04: Scalable to 100+ users
- NFR-05: Compliant with healthcare data regulations

8. Assumptions

- Webcams/mobile devices available
- Basic internet connectivity

- Doctors are digitally literate

9. Milestones

- Week 1: UI Design, Schema Planning
- Week 2-3: Registration & Face Capture
- Week 4: Backend & API Setup
- Week 5: Face Recognition Integration
- Week 6-7: Record Management
- Week 8: Dashboards & Roles
- Week 9-10: Testing & Deployment

10. KPIs (Success Metrics)

- Identification success rate > 95%
- 0 unauthorized access attempts
- Number of patients registered
- Number of records uploaded
- User satisfaction > 85%

11. Future Enhancements

- Patient mobile app
- QR/fingerprint fallback
- AI-based suggestions
- E-prescription integration
- Appointment booking